

Venkata Gopi Ranjith Kumar Maturi

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Profile Summary

- Data Engineer with 5+ years of experience designing scalable pipelines across healthcare, CPG, telecom, and banking domains in US and Indian markets.
- Data Engineer designing HIPAA-compliant data pipelines at HCA Healthcare to unify EHR, clinical trial, and IoT data, enabling real-time patient care analytics and accelerating FDA submissions via Azure Synapse and Databricks.
- Data Engineer at PepsiCo optimizing FMCG supply chains by processing IoT sensors and sales data via Azure Data Factory and Databricks, enhancing demand forecasting and scalability for 100+ global plants.
- Data Engineer at Verizon via TCS Migrated legacy Hadoop/Spark batch pipelines to cloud-native workflows. Orchestrated telecom usage analytics with Airflow.
- Junior Data Engineer at Lloyds Banking Group via TCS Modernized RBI-compliant financial reporting by transitioning mainframe data to Hadoop/Hive. Scripted audit utilities in Shell/Python.
- Proficient in Azure, AWS (Redshift, Glue, Kinesis), Hadoop ecosystem (Hive, Spark), and IaC (Terraform).
- Domain expertise in regulatory standards: HIPAA (healthcare), RBI (banking), and telecom data governance.
- Led end-to-end modernization projects from legacy systems (mainframe) to cloud-native architectures.

Skills

- **Languages:** Python, Shell, SQL, HiveQL, Scala, R, COBOL
- **Cloud Platforms:** Azure (Data Factory, Synapse Analytics, Databricks, Stream Analytics, Event Hubs, Blob Storage, Cosmos DB, Key Vault, Logic Apps, Purview, Azure Analysis Services, Azure SQL Database), AWS (S3, EMR, Redshift, Glue, Kinesis, Athena)
- **Big Data & Analytics:** Hadoop, Hive, Spark (PySpark), Delta Lake, Parquet, Databricks, Airflow, Kafka, Snowflake
- **Orchestration & Automation:** Apache Airflow, Jenkins, Terraform, Helm, Power Automate
- **Data Modeling & Governance:** Azure Purview, Glue Data Catalog, RBAC, GDPR, HIPAA, SOX, RBI, PRA, 3GPP compliance
- **Visualization & Reporting:** Power BI, Azure Analysis Services
- **DevOps & CI/CD:** Docker, Kubernetes (EKS), Terraform, Jenkins
- **Security:** SAS Tokens, Azure AD Authentication, Azure Key Vault, Private Endpoints, PII masking
- **Other Tools & Technologies:** VSAM, FTP, SQL Server, Mainframe, SAP ERP, MRI & IoT Data Systems

Relevant Experience

HCA Healthcare

Jun 2024 - Present

Data Engineer

Dallas, TX

- Designed and deployed scalable Azure data lake pipelines to consolidate EHR and clinical trial data, enabling real-time analytics for patient care optimization. Integrated HIPAA-compliant encryption and RBAC for secure access to PHI.
- Architected azure synapse analytics workflows to unify structured/unstructured data from MRI and lab systems, improving diagnostic reporting accuracy for oncology teams.
- Automated legacy ETL migration from on-premises SQL server to azure data factory, reducing manual effort in data integration by 35% for critical care datasets.
- Streamlined ICU sensor data ingestion using Azure stream analytics and event hubs, enhancing real-time monitoring dashboards for emergency response teams.
- Implemented Delta Lake schemas in Azure Databricks to standardize clinical trial data, accelerating FDA submission timelines for new drug approvals.
- Secured data pipelines with Azure key vault and private endpoints, ensuring compliance with HIPAA and GDPR for cross-border patient data.
- Collaborated with clinicians to design azure purview-driven metadata frameworks, improving traceability of chemotherapy treatment datasets.
- Optimized azure SQL database performance via indexing and partitioning, reducing query latency for high volume billing and claims analytics.

PepsiCo

Jul 2023 - May 2024

Data Engineer

Dallas, TX

- Built Azure data factory pipelines to ingest and transform IOT sensor data from 100+ manufacturing plants, optimizing supply chain demand forecasting.
- Designed Azure Databricks workflows to process terabytes of sales data, enabling AI-driven insights for regional inventory replenishment strategies.
- Migrated legacy SAP ERP datasets to Azure Synapse, reducing latency in financial reporting and enhancing decision-making for North American sales operations.
- Automated retail partner data onboarding using Azure logic apps and power automate, cutting manual validation efforts by 40%.
- Implemented Azure stream analytics for real-time production line monitoring, reducing equipment down time through predictive maintenance alerts.
- Secured Azure blob storage with SAS tokens and Azure ad authentication, ensuring compliance with CPG data governance standards.
- Developed power BI dashboards using Azure analysis services to visualize SKU performance, supporting pricing strategy adjustments.

- Optimized Azure cosmos DB throughput for global distributor data, improving scalability for peak season transaction processing.

Verizon

Jun 2020 - Dec 2021

Data Engineer

Hyderabad, India

- Migrated Hadoop CDR pipelines to AWS EMR for processing telecom records. Optimized Spark jobs using Kryo serialization.
- Ingested telecom log and call detail records (CDRs) into Amazon S3 buckets for cost-effective storage and downstream Spark processing on AWS EMR.
- Configured Amazon S3 lifecycle rules for log archiving and automated cleanup, reducing storage costs by 25%.
- Built Airflow DAGs to orchestrate 5G network usage workflows. Reduced pipeline failures through SLA based monitoring.
- Automated EMR cluster provisioning using Terraform and Jenkins. Accelerated deployment times for development teams.
- Designed Parquet partitioning for historical call datasets. Enhanced Athena query performance for billing analytics.
- Developed PySpark validators to flag missing tower data in streams and integrated findings with NOC dashboards via Kafka.
- Modernized COBOL batch jobs to Spark SQL for roaming charges. Eliminated manual reconciliation steps.
- Containerized legacy Java parsers using Docker and EKS. Enabled scalable deployments with Helm charts.
- Collaborated with engineers to align data models with 3GPP standards. Streamlined VoLTE call drop analytics.
- Documented GDPR-compliant data lineage using Glue Catalog and audited PII masking in customer reports, ensuring data privacy and compliance with regulations.

Llyods Banking Group

Mar 2018 - May 2020

Junior Data Engineer

Hyderabad, India

- Modernized core banking reports by migrating COBOL mainframe data to Hadoop. Developed HiveQL transformations for regulatory submissions.
- Scripted Python validators to ensure accuracy of migrated transaction records. Automated reconciliation of legacy VSAM datasets.
- Designed Shell utilities to automate RBI-compliant audit documentation. Collaborated with compliance teams on control frameworks.
- Transitioned batch reporting systems to Hive for capital adequacy calculations. Aligned workflows with PRA guidelines.
- Built data quality checks for loan portfolio datasets. Supported risk management teams in anomaly detection.
- Migrated legacy flat-file archives to HDFS for scalable storage. Improved accessibility for fraud detection analytics.
- Documented end-to-end data lineage for SOX compliance. Coordinated with auditors for control validations.
- Automated file transfers from mainframe to Hadoop using FTP scripts. Reduced manual intervention in daily ETL jobs.

Education

University of Texas at Dallas

Jan 2022 - May 2023

Masters, Business Analytics

GPA: 3.73/4

V R Siddhartha Engineering College

Jun 2013 - Mar 2017

Bachelors, Electronics and Instrumentation Engineering

GPA: 7/10

Certifications

- AWS Certified Data Analytics - Specialty
- Microsoft Certified: Azure Fundamentals (AZ-900)